UNIVERSITY OF TORONTO
Faculty of Arts & Science

April 2025 Examination

CSC 165 H1S
Math. Expression & Reasoning for C.S.
Instructor(s): G. Baumgartner & F. Pitt

Duration: 3 hours

Aids Allowed: None
(in particular, NO calculator)

See the notes placed throughout this exam for additional context on each question, which ones are relevant, and which ones are not.
Note: in 2025, implication (→) was represented as "⇒" and biconditional (↔) was represented as "⇔".
Also note: we do NOT have solutions to these questions.

*Do **not** turn this page until you have received the signal to start.*

Write your name and student number below **(do this now)**, then read the instructions.

STUDENTS MUST HAND IN ALL EXAM MATERIALS AT THE END

Crowdmark Automated Matching Area

*Any content within this area (and a small margin around it) will be **overwritten** by Crowdmark.*

- As a student, you help create a fair and inclusive writing environment. If you possess an unauthorized aid during the exam, you may be charged with an academic offence.
- Turn off and place all cell phones, smart watches, electronic devices, and unauthorized study materials in your bag under your desk. If it is left in your pocket, it may be an academic offence.
- When you are done your exam, raise your hand for someone to come and collect your exam. *Do not collect your bag and jacket before your exam is handed in.* If you are feeling ill and unable to finish your exam, *please bring it to the attention of the Exam Facilitator* so it can be recorded before leaving the room. In the event of a fire alarm, do **not** check your cell phone when escorted outside.
- *In order to pass the course, you must earn at least **25%** (17/68) on this exam.*
- *Please use a **dark** pen or pencil* to ensure your answers are legible (even when scanned).
- Answer each question directly on the exam paper, in the space provided. Use "blank" pages for rough work, or for answers that did not fit elsewhere—*indicate clearly the pages that should be marked.*
- This exam contains 7 questions on 24 pages (including 5 "blank" pages, and a Reference Sheet on the last 2 pages), printed on both sides of the paper. *When you start, please make sure that your copy is complete.*

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Total
6	10	9	10	15	11	7	68

Good Luck!

This is a good practice problem about reading, understanding, and instantiating definitions. It will also make you practice thinking about number representations. You are NOT expected to know already how to answer it, but you ARE expected to be able to work it out (though it may take you a bit longer than it would have taken last year's students).

Question 1. [6 marks] Number Representations

In general, positional number representations can combine an integer *and* a fractional part (for example, $16.5 = 1 \cdot 10^1 + 6 \cdot 10^0 + 5 \cdot 10^{-1}$). In binary, for each $m, n \in \mathbb{Z}$ with $m \leq 0 \leq n$, and each $b_m, \ldots, b_0, \ldots, b_{n-1} \in \{0, 1\}$, we define $(b_{n-1} \cdots b_0 \,.\, b_{-1} \cdots b_m)_2$ to be a representation of the number $\sum_{i=m}^{n-1} b_i \cdot 2^i$.

Part (a) [2 marks]

For this part, you may find (some of) the following values useful: $2^{-1} = \frac{1}{2} = .5$, $2^{-2} = \frac{1}{4} = .25$, $2^{-3} = \frac{1}{8} = .125$.

i– Write the expanded summation (first term + second term + $\cdots$ + last term) with each term fully instantiated, for the representation $(110\,.\,011)_2$, then calculate its total (decimal) value.

ii– Give a binary representation for the number 10.75 .

Question 1. (continued)

Part (b) [2 marks]

Let $m \in \mathbb{Z}^-$ and $b_m, \ldots, b_{-1} \in \{0, 1\}$ and $k \in \mathbb{N}$.

Give an expression for the value of $(\,.\,\underbrace{0 \cdots 0}_{k}\, b_{-1} \cdots b_m)_2$, in terms of $(\,.\,b_{-1} \cdots b_m)_2$ and 2^k.

Part (c) [2 marks]

Let $x \in \mathbb{Q}$ and $n \in \mathbb{N}^+$ and $k \in \mathbb{N}$, with $k \leq n$, and $(b_{n-1} \cdots b_k \,.\, b_{k-1} \cdots b_0)_2$ be a binary representation of x.

Show that $2^k \cdot x = (b_{n-1} \cdots b_0)_2$.

IGNORE Part (a): we did not cover GCD (greatest common divisor) this term.
The other two parts are reasonable practice problems.

Question 2. [10 marks] Basic Proofs

In this question, *use only the definitions and properties of gcd, floor, and ceiling provided on the Reference Sheet and given in the question*—you must **prove** any other result about these functions that you want to use in your answer.

Part (a) [2 marks]

Prove that $\forall n \in \mathbb{N},\ n \geq 2 \Rightarrow \gcd(n, n-1) = 1$.

Part (b) [4 marks]

Prove that $\forall x \in \mathbb{R},\ \lceil x + 165 \rceil = \lceil x \rceil + 165$.

In addition to the definitions and properties on the Reference Sheet, you may use the fact that ceil-char is a *characterization*, in other words, it is also true that floor-uniq: $\forall z \in \mathbb{Z}, x \in \mathbb{R},\ x \leq z < x + 1 \Rightarrow z = \lceil x \rceil$.

Question 2. (continued)

Part (c) [4 marks]

Prove that $\forall n \in \mathbb{N}, \left\lfloor \frac{\lfloor n/2 \rfloor}{2} \right\rfloor = \left\lfloor \frac{n}{4} \right\rfloor$.

Hints. Introduce appropriate cases (possibly twice). You may simplify floors of concrete numbers directly (e.g., $\lfloor 1.65 \rfloor = 1$). You may use the following additional property (but you are **not** obliged to do so).

floor-int: $\forall x \in \mathbb{R}, k \in \mathbb{Z}, \lfloor x + k \rfloor = \lfloor x \rfloor + k$

"f ∈ Ω(g)" means: f ≳ g. Answer the question directly from the definitions. We recommend you practice using a "compressed" eventually style proof, but this is not required. If you are not able to answer Part (b), then it means you do NOT understand your own proof. Review the role of universal assumptions and ask us questions!

Question 3. [9 marks] Asymptotics

Part (a) [7 marks]

Prove that:

$$\forall f, g, h, k : \mathbb{N} \to \mathbb{R}^{\geq 0},\ f \in \Omega(g) \wedge h \in \omega(k) \Rightarrow f \cdot h \in \omega(g \cdot k).$$

(Use **only** the Asymptotic Definitions provided on the Reference Sheet, ***not*** *any of the Asymptotic Hierarchy or Asymptotic Properties*; in particular, you **cannot** use prod-ω.)

Question 3. (continued)

Part (a) (continued)

Part (b) [2 marks]

Let $f, g, h, k : \mathbb{N} \to \mathbb{R}^{\geq 0}$. Suppose $c = 3, n_0 = 2$ witnesses $f \in \Omega(g)$. To prove instance $c = 5$ of $f \cdot h \in \omega(g \cdot k)$, which instance of $h \in \omega(k)$ is used in *your* proof (from the previous part)?

This should look suspiciously familiar (see ES13)... But Part (b) is slightly different and worth trying, after you have solved the corresponding problem on ES13.

Question 4. [10 marks] Induction

Part (a) [7 marks]

Use Simple Induction to prove that $\forall a, b \in \mathbb{N}, a < b \Rightarrow \forall n \in \mathbb{N}, (b - a) \mid (b^n - a^n)$. Write a detailed proof structure, using the conventions from the course. In particular, label each component of your proof with its standard name ("Base Case", "Inductive Step", "Inductive Hypothesis"), and state explicitly the statement you are attempting to prove at each step.

Hints. Choose carefully which variable to do induction on. For this problem, use the convention that $0^0 = 1$.

Question 4. (continued)

Part (b) [3 marks]

Fill in each blank below with an expression or a concrete value. Each answer must be based on *your* proof from the previous part.

i– What witness does *your* proof generate for $(b - a) \mid (b^0 - a^0)$? _____

ii– What witness does *your* proof generate for $(b - a) \mid (b^1 - a^1)$? _____

iii– What witness does *your* proof generate for $(b - a) \mid (b^2 - a^2)$? _____

iv– What witness does *your* proof generate for $(b - a) \mid (b^3 - a^3)$? _____

Show the calculation performed by *your* proof to obtain this answer.

This entire question is relevant to us. Careful! We are NOT saying that we will definitely have a question like this on our own exam! But each part asks something that we could ask.

Question 5. [15 marks] Logic & Symbolic Expressions

Part (a) [1 mark]

Let D be a non-empty domain and $P, Q, R : D \to \mathbb{B}$, and consider the following statement:

$$\forall v \in D,\ P(v) \Rightarrow (Q(v) \Rightarrow R(v)).$$

Fill in the body of the following so that it is **equivalent** to the statement above, using conjunction and **only one** implication.

$$\forall v \in D,\ \underline{\hspace{6cm}}$$

Part (b) [4 marks]

Let D be a non-empty domain and suppose P and Q are predicates on D.

For each of the following statements, indicate whether it is true, is false, or depends on P and/or Q, by drawing a circle or a box around the appropriate answer. *No justification is required*.

i– $\forall v \in D,\ P(v) \wedge Q(v) \Rightarrow P(v)$: is true is false depends on P and/or Q

ii– $\forall v \in D,\ P(v) \vee Q(v) \Rightarrow P(v)$: is true is false depends on P and/or Q

iii– $\forall v \in D,\ P(v) \Rightarrow P(v) \wedge Q(v)$: is true is false depends on P and/or Q

iv– $\forall v \in D,\ P(v) \Rightarrow P(v) \vee Q(v)$: is true is false depends on P and/or Q

Question 5. (continued)

Part (c) [3 marks]

Consider the following statement:

$$\exists y \in \mathbb{N}, \forall x \in \mathbb{N}, y < x.$$

Expand the outer (existential) quantification. Show at least 3 instances.

Give a negation of the statement, in a form that has **no** explicit negation.

State whether the original statement is true or false. *No justification is required.*

Part (d) [2 marks]

Consider the following statement (pay attention to the domains):

$$\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, y < x.$$

Expand the outer (universal) quantification. Show at least 3 instances.

State whether the statement is true or false. *No justification is required.*

Question 5. (continued)

Part (e) [1 mark]

State whether the following statement is true or false (pay attention to the domains). *No justification is required.*

$$\exists y \in \mathbb{Z}, \forall x \in \mathbb{N}, y < x.$$

Part (f) [4 marks]

Let $n \in \mathbb{N}$, $a_0, a_1, \ldots, a_n \in \mathbb{N}$, and $b_0, b_1, \ldots, b_n \in \mathbb{N}$. Let $I = \{0, 1, \ldots, n\}$. Define predicates E and O on $\mathbb{N}$, as follows: for each $m \in \mathbb{N}$, $E(m)$ means "m is even", and $O(m)$ means "m is odd".

In your answers to this question, use **only** conjunction, disjunction, quantification with domain I, and/or predicates E and O applied to individual elements from $a_0, \ldots, a_n, b_0, \ldots, b_n$. *Do **not** use multiplication nor addition.*

Hint. For one (or both) of the questions, you might find it easier to first express the negation of the original statement, and then negate that.

Consider the statement:

$$\left(\prod_{i=0}^{n} a_i\right) + \left(\prod_{i=0}^{n} b_i\right) \quad \text{is odd.}$$

i– Write a an equivalent symbolic logical statement, in the form of a conjunction (of necessary conditions).

ii– Write a an equivalent symbolic logical statement, in the form of a disjunction (of sufficient conditions).

This is very similar to the last question on Exercise Set 12 (algorithm "maxfact"), except with an additional twist based on GCDs. Since we did NOT study GCD this term, the particular algorithm on this exam is NOT one we expect you to be able to analyze. But we do expect you to be able to analyze similar algorithms (as on ES12).

Question 6. [11 marks] Worst-Case Runtime

Consider the following algorithm. (Recall that "%" is the *modulus* operator in Python: `x % y` evaluates to the remainder when `x` is divided by `y`; in particular, `x % y == 0` iff `y | x`.)

```
0  def max_gcd(A: list[int]) -> int:
1      """ Precondition: A is a non-empty list of integers, and
2          ∀ i ∈ ℕ, i < len(A) ⇒ 1 ≤ A[i] ≤ len(A). """
3      m = 1
4      for i in range(len(A)):       # Loop 1 (i = 0,...,n-1)
5          for j in range(i):        # Loop 2 (j = 0,...,i-1)
6              g = min(A[i], A[j])
7              while not (A[j] % g == 0 and A[i] % g == 0):  # Loop 3
8                  g = g - 1
9              if g > m:  m = g
10     return m
```

Part (a) [2 marks]

Let $\mathcal{I}$ be the set of all inputs that satisfy `max_gcd`'s precondition. For each input $A \in \mathcal{I}$, define $RT(A)$ to be the *total* number of times that line 7 runs (*including when the condition evaluates to false*), during the call `max_gcd`(A).

List *every* element $A \in \mathcal{I}_2$, and determine the value of $RT(A)$ for each A.

Question 6. (continued)

Part (b) [9 marks]

Prove a *tight* bound on WC, the worst-case running time function for algorithm `max_gcd`, by finding a simple function $f : \mathbb{N} \to \mathbb{R}^{\geq 0}$ such that $WC \in \Theta(f)$, and proving this Θ relationship. Use the method demonstrated on our Worksheets and Problem Sets. In particular, use the definitions and/or properties from the Reference Sheet to simplify your final answer.

Hint. $\forall n \in \mathbb{N},\ n \geq 2 \Rightarrow \gcd(n, n-1) = 1$.

Question 6. (continued)

Part (b) (continued)

In the definition of the runtime function RT, you may substitute "lines" for "steps".
Answer Parts (d) and (e) for algorithm maxfact from ES12.

Question 7. [7 marks] Best-Case Runtime

Recall that for each program p, we define $\mathcal{I} = \{\text{every input that satisfies } p\text{'s preconditions}\}$, and for each $n \in \mathbb{N}$, we define $\mathcal{I}_n = \{x \in \mathcal{I} : \text{size}(x) = n\}$. Also, we define the *runtime function* $RT : \mathcal{I} \to \mathbb{R}^{\geq 0}$ as follows: for each $x \in \mathcal{I}$, $RT(x)$ is the total number of "steps" performed during the call $p(x)$. With this context:

- the *worst-case* runtime $WC : \mathbb{N} \to \mathbb{R}^{\geq 0}$ is defined to be, for each $n \in \mathbb{N}$: $WC(n) = \max\{RT(x) : x \in \mathcal{I}_n\}$ *(this is the definition we have been using)*;
- the *best-case* runtime $BC : \mathbb{N} \to \mathbb{R}^{\geq 0}$ is defined to be, for each $n \in \mathbb{N}$: $BC(n) = \min\{RT(x) : x \in \mathcal{I}_n\}$ *(this is a new definition)*.

Part (a) [1 mark]

Let p be a program with input set $\mathcal{I}$. Let $x \in \mathcal{I}_7$ and suppose $RT(x) = 165$.

i– What bound can you conclude on $WC(7)$?

ii– What bound can you conclude on $BC(7)$?

Part (b) [1 mark]

For each set $S \subseteq \mathbb{N}$ with $S \neq \emptyset$, define "$\min(S)$" to be the *smallest* element of S. Let $B \in \mathbb{R}$.

Write a symbolic logical expression that is equivalent to "$B \leq \min(S)$". *Do* ***not*** *use* min, *and for any quantification, use only domain S or* $\mathbb{N}$.

Part (c) [1 mark]

For each set $S \subseteq \mathbb{N}$ with $S \neq \emptyset$, define "$\min(S)$" to be the *smallest* element of S. Let $B \in \mathbb{R}$.

Write a symbolic logical expression that is equivalent to "$B \geq \min(S)$". *Do* ***not*** *use* min, *and for any quantification, use only domain S or* $\mathbb{N}$.

Question 7. (continued)

Part (d) [2 marks]

Give an input sequence (also called "input family") $A_0 \in \mathcal{I}_0, A_1 \in \mathcal{I}_1, A_2 \in \mathcal{I}_2, \ldots, A_n \in \mathcal{I}_n, \ldots$ for algorithm `max_gcd` from Question 6, such that $RT(A_n) \in \Theta(BC(n))$.

Part (e) [2 marks]

State a simple function $f : \mathbb{N} \to \mathbb{R}^{\geq 0}$ such that $RT(A_n) \in \Theta(f(n))$. *No proof is required.*

Does the existence of this input family prove that $BC \in \mathcal{O}(f)$ or that $BC \in \Omega(f)$?

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Final Examination

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April 2025

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Final Examination

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Reference Sheet

*You may detach this sheet but **you must hand it in with your exam**.*

Our own reference sheet will look quite different from this, and be much closer to the reference sheets from our term tests. Our ref. sheet will definitely contain asymptotic definitions and properties, and likely other elements (details to be determined and posted separately).

General Definitions and Properties.

div-def: For each $n, d \in \mathbb{Z}$, "$d \mid n$" means: $\exists b \in \mathbb{Z},\ n = bd$

div-bound: $\forall n, d \in \mathbb{N}^+,\ d \mid n \Rightarrow d \leq n$

div-combo: $\forall m, n, d, s, t \in \mathbb{Z},\ d \mid m \wedge d \mid n \Rightarrow d \mid sm + tn$

gcd-def: $\forall m, n \in \mathbb{N}^+,\ \gcd(m, n) \in \mathbb{N}^+ \wedge \gcd(m, n) \mid m \wedge \gcd(m, n) \mid n \wedge (\forall d \in \mathbb{N}^+,\ d \mid m \wedge d \mid n \Rightarrow d \leq \gcd(m, n))$

gcd-uniq: $\forall m, n, g \in \mathbb{N}^+,\ g \mid m \wedge g \mid n \wedge (\forall d \in \mathbb{N}^+,\ d \mid m \wedge d \mid n \Rightarrow d \leq g) \Rightarrow g = \gcd(m, n)$

prime-def: For each $n \in \mathbb{N}^+$, "$Prime(n)$" means: $n > 1 \wedge \forall d \in \mathbb{N}^+,\ d \mid n \Rightarrow d = 1 \vee d = n$

prime-fact: $\forall n \in \mathbb{N}^+,\ n > 1 \wedge \neg Prime(n) \Rightarrow \exists a, b \in \mathbb{N}^+,\ 1 < a < n \wedge 1 < b < n \wedge n = ab$

qrt: $\forall n \in \mathbb{N}, d \in \mathbb{N}^+, \exists q, r \in \mathbb{N},\ r < d \wedge n = dq + r$ (and q, r are unique)

floor-def: For each $x \in \mathbb{R}$, "$\lfloor x \rfloor$" means: $\max\{z \in \mathbb{Z} : z \leq x\}$

floor-char: $\forall x \in \mathbb{R},\ \lfloor x \rfloor \in \mathbb{Z} \wedge x - 1 < \lfloor x \rfloor \leq x$

ceil-def: For each $x \in \mathbb{R}$, "$\lceil x \rceil$" means: $\min\{z \in \mathbb{Z} : z \geq x\}$

ceil-char: $\forall x \in \mathbb{R},\ \lceil x \rceil \in \mathbb{Z} \wedge x \leq \lceil x \rceil < x + 1$

bin-def-int: For each $k \in \mathbb{N}$, $b_0, \ldots, b_{k-1} \in \{0, 1\}$, "$(b_{k-1} \cdots b_0)_2$" means: $\sum_{i=0}^{k-1}(b_i \cdot 2^i)$

bin-def-frac: For each $k \in \mathbb{N}$, $b_1, \ldots, b_k \in \{0, 1\}$, "$(.b_1 \cdots b_k)_2$" means: $\sum_{i=1}^{k}(b_i \cdot 2^{-i})$

bin-equiv: $\forall k \in \mathbb{N}, b_1, \ldots, b_k \in \{0, 1\},\ (.b_1 \cdots b_k)_2 = (b_1 \ldots b_k)_2/2^k$

Reference Sheet

*You may detach this sheet but **you must hand it in with your exam**.*

Asymptotic Definitions. For each $f, g : \mathbb{N} \to \mathbb{R}^{\geq 0}$:

def-ω: "$f \in \omega(g)$" means: $\forall c \in \mathbb{R}^+, \exists n_0 \in \mathbb{R}^{\geq 0}, \forall n \in \mathbb{N}, n \geq n_0 \Rightarrow f(n) \geq c \cdot g(n)$

def-$\mathcal{O}$: "$f \in \mathcal{O}(g)$" means: $\exists c \in \mathbb{R}^+, n_0 \in \mathbb{N}, \forall n \in \mathbb{N}, n \geq n_0 \Rightarrow f(n) \leq c \cdot g(n)$

def-Ω: "$f \in \Omega(g)$" means: $g \in \mathcal{O}(f)$

def-Θ: "$f \in \Theta(g)$" means: $f \in \mathcal{O}(g) \wedge f \in \Omega(g) \quad \big(\equiv f \in \mathcal{O}(g) \wedge g \in \mathcal{O}(f)\big)$

def-dom: "g eventually dominates f" means: $\exists n_0 \in \mathbb{N}, \forall n \in \mathbb{N}, n \geq n_0 \Rightarrow f(n) \leq g(n)$
(in particular, if g eventually dominates f, then $f \in \mathcal{O}(g)$)

Asymptotic Hierarchy. For each $a, b \in \mathbb{R}^+$, $c, d \in \mathbb{R}^{>1}$ with $a < b$ and $c < d$:

con-con: $a \in \Theta(b) \wedge b \in \Theta(a)$

con-log: $\log n \in \omega(1) \wedge 1 \in \mathcal{O}(\log n) \wedge \log n \notin \mathcal{O}(1)$

con-pol: $n^a \in \omega(1) \wedge 1 \in \mathcal{O}(n^a) \wedge n^a \notin \mathcal{O}(1)$

con-exp: $c^n \in \omega(1) \wedge 1 \in \mathcal{O}(c^n) \wedge c^n \notin \mathcal{O}(1)$

log-log: $\log_c n \in \Theta(\log_d n) \wedge \log_d n \in \Theta(\log_c n)$

log-pol: $n^a \in \omega(\log n) \wedge \log n \in \mathcal{O}(n^a) \wedge n^a \notin \mathcal{O}(\log n)$

log-exp: $c^n \in \omega(\log n) \wedge \log n \in \mathcal{O}(c^n) \wedge c^n \notin \mathcal{O}(\log n)$

pol-pol: $n^b \in \omega(n^a) \wedge n^a \in \mathcal{O}(n^b) \wedge n^b \notin \mathcal{O}(n^a)$

pol-exp: $c^n \in \omega(n^a) \wedge n^a \in \mathcal{O}(c^n) \wedge c^n \notin \mathcal{O}(n^a)$

exp-exp: $d^n \in \omega(c^n) \wedge c^n \in \mathcal{O}(d^n) \wedge d^n \notin \mathcal{O}(c^n)$

Asymptotic Properties. For each $a \in \mathbb{R}^{\geq 0}$, $b \in \mathbb{R}^+$, $f, g, h, k : \mathbb{N} \to \mathbb{R}^{\geq 0}$ (*eventually* defined and non-negative):

symmetry: $f \in \Theta(g) \Leftrightarrow g \in \Theta(f)$

reflexivity: $b \cdot f \in \Theta(f)$ (in particular, $f \in \Theta(f)$)

transitivity-ω: $f \in \omega(g) \wedge g \in \omega(h) \Rightarrow f \in \omega(h)$
(*and the same with $\mathcal{O}$ or Ω or Θ in place of ω*)

ω-iff-R: $f \in \Theta(g) \Rightarrow (h \in \omega(f) \Leftrightarrow h \in \omega(g))$
(*and the same with $\mathcal{O}$ or Ω or Θ in place of ω*)

ω-iff-L: $f \in \Theta(g) \Rightarrow (f \in \omega(h) \Leftrightarrow g \in \omega(h))$
(*and the same with $\mathcal{O}$ or Ω or Θ in place of ω*)

ω-Ω: $f \in \omega(g) \Rightarrow f \in \Omega(g)$

ω-$\mathcal{O}$: $f \in \omega(g) \wedge g$ is eventually positive $\Rightarrow f \notin \mathcal{O}(g)$

sum-poly: $\sum_{i=0}^{n} i^a \in \Theta(n^{a+1})$

sum-ω: $g \in \mathcal{O}(k) \wedge h \in \omega(k) \Rightarrow f + h \in \omega(g + k)$

sum-$\mathcal{O}$: $f \in \mathcal{O}(g) \wedge h \in \mathcal{O}(k) \Rightarrow f + h \in \mathcal{O}(g + k)$
(*and the same with Ω or Θ in place of $\mathcal{O}$*)

prod-ω: $f \in \Omega(g) \wedge h \in \omega(k) \Rightarrow f \cdot h \in \omega(g \cdot k)$

prod-$\mathcal{O}$: $f \in \mathcal{O}(g) \wedge h \in \mathcal{O}(k) \Rightarrow f \cdot h \in \mathcal{O}(g \cdot k)$
(*and the same with Ω or Θ in place of $\mathcal{O}$*)

add-Ω: $f + g \in \Omega(f)$

add-$\mathcal{O}$: $h \in \Omega(g) \wedge f \in \mathcal{O}(h) \Rightarrow f + g \in \mathcal{O}(h)$
(*and the same with Θ in place of $\mathcal{O}$*)

sub-Ω: $f \in \omega(g) \wedge f \in \Omega(h) \Rightarrow f - g \in \Omega(h)$
(*and the same with Θ in place of Ω*)

sub-$\mathcal{O}$: $f - g \in \mathcal{O}(f)$

sum-bound: f, g eventually dominate $1 \wedge f \in \Theta(g) \Rightarrow \sum_{i=0}^{n} f(i) \in \Theta\Big(\sum_{i=0}^{n} g(i)\Big)$

sum-comp: $h \in \omega(1) \wedge \sum_{i=0}^{n} f(i) \in \Theta(g(n)) \Rightarrow \sum_{i=0}^{h(n)} f(i) \in \Theta\Big(g(h(n))\Big)$

comp-poly: $h \in \omega(1) \wedge h \in \Theta(k) \wedge g(n) \in \Theta(n^a) \Rightarrow g(h(n)) \in \Theta\Big((k(n))^a\Big)$

ceil-Θ: f eventually dominates $1 \Rightarrow \lceil f + a \rceil \in \Theta(f)$; in particular, f eventually dominates $1 \Rightarrow \lceil f \rceil \in \Theta(f)$

floor-Θ: f eventually dominates $1 \Rightarrow \lfloor f + a \rfloor \in \Theta(f)$; in particular, f eventually dominates $1 \Rightarrow \lfloor f \rfloor \in \Theta(f)$

Total Marks = 68